

5G RAN

Ultimate Cloud X Experience Feature Parameter Description

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Date	2025-12-31



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Contents

1 Change History.....1
1.1 5G RAN10.1 Draft A (2025-12-31).....1

2 About This Document.....4
2.1 General Statements.....4
2.2 Features in This Document.....5
2.3 Differences Between NSA and SA.....5
2.4 Differences Between High Frequency Bands and Low Frequency Bands.....5

3 Overview.....6

4 Cloud X Service Experience Improvement.....7
4.1 Principles.....7
4.1.1 UE Identification.....7
4.1.1.1 Cloud X UE Identification.....7
4.1.2 Scheduling Guarantee.....8
4.1.2.1 Cloud X Services.....8
4.1.2.2 Cloud X High-Priority UE Guarantee.....10
4.1.2.3 Predicted Uplink Scheduling for Cloud X Services.....11
4.1.2.4 QCI-specific Scheduling Based on the Downlink Target IBLER.....12
4.1.2.5 Downlink MCS Adjustment Optimization.....12
4.1.2.6 Explicit Scheduling for Downlink Second Retransmission.....12
4.1.3 Mobility Management.....14
4.1.3.1 Integrity of Cloud X Frame Transmission During Handovers.....14
4.1.4 Carrier Management.....14
4.1.4.1 Intra-Base-Station CA Downlink Data Split for Cloud X Services.....14
4.1.4.2 Latency-based Optimal Carrier Selection.....15
4.1.5 Other Functions.....16
4.1.5.1 Uplink ROHC-TCP Compression.....16
4.2 Network Analysis.....16
4.2.1 Benefits.....16
4.2.2 Impacts.....17
4.3 Requirements.....17
4.3.1 Licenses.....17
4.3.2 Functions.....18

4.3.2.1 Prerequisite Functions.....	18
4.3.2.2 Mutually Exclusive Functions.....	21
4.3.2.3 Function Impacts.....	22
4.3.3 Hardware.....	27
4.3.4 Others.....	28
4.4 Operation and Maintenance.....	28
4.4.1 Data Configuration.....	28
4.4.1.1 Data Preparation.....	28
4.4.1.2 Using MML Commands.....	33
4.4.1.3 Using the MAE-Deployment.....	35
4.4.2 Activation Verification.....	35
4.4.3 Network Monitoring.....	37
5 Parameters.....	39
6 Counters.....	41
7 Glossary.....	42
8 Reference Documents.....	43

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-11-07).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between Cloud X UE identification and time-domain power saving BWP. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between Cloud X services and QCI-specific uplink data compression. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added methods of observing QCI-level MAC- and RLC-layer delay for Cloud X functions in NSA networking. For details, see 4.4.2 Activation Verification .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Reorganized this document and reworded the information.

- The editorial changes that involve changes in meanings are as follows:
 - Added an E2E transmission diagram of Cloud X services. For details, see [3 Overview](#).
 - Added descriptions of downlink Cloud X service characteristics. For details, see [4.1.2.1 Cloud X Services](#).
 - Revised descriptions of principles of frame delay-based scheduling and added a schematic diagram. For details, see [4.1.2.1 Cloud X Services](#).
 - Moved descriptions of the resource restriction threshold for Cloud X UEs to descriptions of frame delay-based scheduling. For details, see [4.1.2.1 Cloud X Services](#).
 - Revised descriptions of Cloud X high-priority UE guarantee. For details, see [4.1.2.2 Cloud X High-Priority UE Guarantee](#).
 - Added the background and schematic diagram of integrity of Cloud X frame transmission during handovers. For details, see [4.1.3.1 Integrity of Cloud X Frame Transmission During Handovers](#).
 - Added a schematic diagram of intra-base-station CA downlink data split for Cloud X services. For details, see [4.1.4.1 Intra-Base-Station CA Downlink Data Split for Cloud X Services](#).
 - Revised descriptions of UE capability query in uplink ROHC-TCP compression. For details, see [4.1.5.1 Uplink ROHC-TCP Compression](#).
 - Added the background for downlink MCS adjustment optimization and revised descriptions of the principle of this function. For details, see [4.1.2.5 Downlink MCS Adjustment Optimization](#).
 - Added a diagram for explicit scheduling for downlink second retransmission. For details, see [4.1.2.6 Explicit Scheduling for Downlink Second Retransmission](#).
 - Revised descriptions of the MML command examples for functions related to Cloud X service experience improvement. For details, see [4.4.1.2 Using MML Commands](#).

- Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include chapter/section reorganization and text-to-table transformation.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

Feature ID	Feature Name	Chapter/Section
FOFD-081207	Deterministic Latency Guarantee for CloudX	4 Cloud X Service Experience Improvement

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Deterministic Latency Guarantee for CloudX	Uplink ROHC-TCP compression: Supported only in SA networking. Integrity of Cloud X frame transmission during handovers: Supported only in SA networking Latency-based optimal carrier selection: Supported only in SA networking Other functions: None	4 Cloud X Service Experience Improvement

2.4 Differences Between High Frequency Bands and Low Frequency Bands

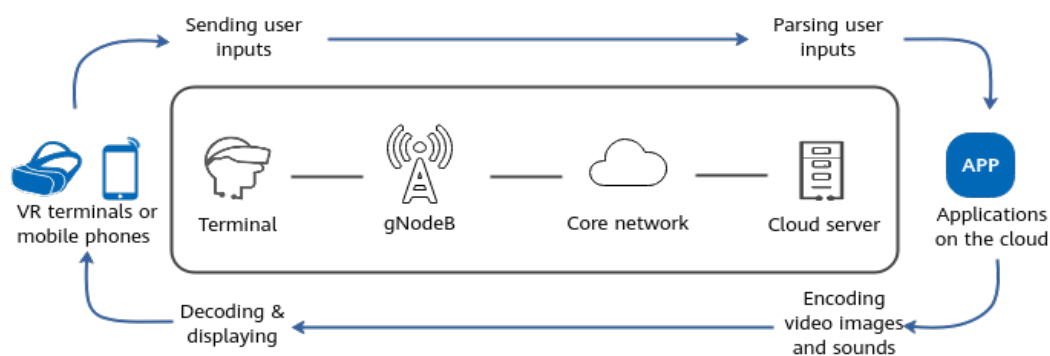
This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Deterministic Latency Guarantee for CloudX	Supported only in low frequency bands	4 Cloud X Service Experience Improvement

3 Overview

Cloud X generally refers to Cloud VR and Cloud Gaming. For those applications, their computing and rendering tasks are transferred to a cloud server. The cloud server encodes the display output and sound output of those applications and transmits the encoded data to terminals (such as VR terminals or mobile phones) in real time over the network. The terminals then decode the data in real time and display the video and audio. In addition, the terminals send user inputs to the cloud server instantly for real-time control. See [Figure 3-1](#).

Figure 3-1 E2E transmission of Cloud X services



To ensure Cloud X service experience, Huawei introduces the ultimate Cloud X experience solution. The gNodeB identifies Cloud X UEs and uses policies such as scheduling guarantee and mobility management to implement differentiated scheduling and transmission.

4 Cloud X Service Experience Improvement

4.1 Principles

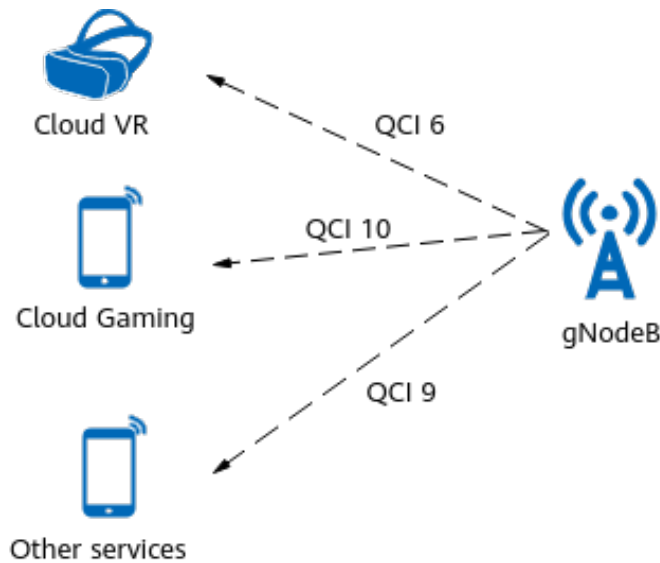
4.1.1 UE Identification

4.1.1.1 Cloud X UE Identification

A critical challenge for Cloud X services is how to distinguish Cloud X UEs from heterogeneous service traffic on the network. To achieve this goal, Huawei introduces the Cloud X UE identification function. Specifically:

1. The core network allocates different 5QIs or QCI to Cloud X UEs and other UEs.
When a radio access bearer (RAB) is set up, the core network notifies the gNodeB of the 5QI or QCI of each UE and the gNodeB maps the 5QIs to the corresponding QCIs.
2. The gNodeB uses the **gNBQciBearer.ServiceType** parameter to configure the service type corresponding to a QCI, as shown in [Figure 4-1](#).
 - When this parameter is set to **EMBB_CLOUD_VR**, the QCI-specific bearer carries Cloud VR services.
 - When this parameter is set to **EMBB_CLOUD_GAMING**, the QCI-specific bearer carries Cloud Gaming services.
3. If a QCI-specific bearer for Cloud VR or Cloud Gaming services is established for a UE, the gNodeB identifies the UE as a Cloud VR or Cloud Gaming UE.

Figure 4-1 Mapping between QCI and service types



NOTE

- After a UE is identified as a Cloud VR or Cloud Gaming UE, if the service type configuration corresponding to the 5QI/QCI on the base station side changes (that is, the **gNBQciBearer.ServiceType** configuration changes) or the 5QI/QCI configured on the core network changes, the UE needs to re-access the network. Otherwise, the user attribute does not change.
- When multiple types of bearers (Cloud VR or Cloud Gaming bearers) are set up for a UE at the same time, only the bearer that is set up first takes effect.
- It is recommended that Cloud X services use dedicated bearers to improve Cloud X service experience and mitigate the impact on other services.
- If the gNodeB identifies that a UE has a bearer in 3GPP Release 18 PDU set mode, some functions in Cloud X service experience improvement can also take effect. For details about the functions that can take effect and their effective mechanisms, see descriptions in "Other PDU Set-based Downlink Experience Enhancements" in *Layered QoS*.

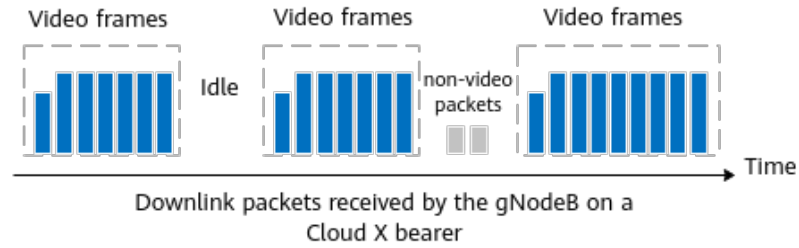
4.1.2 Scheduling Guarantee

4.1.2.1 Cloud X Services

Cloud X UEs have strict requirements on delay and rate. When the **CLOUD_VR_SERVICE_SWITCH** option of the **NRCCellAlgoSwitch.CloudVrSwitch** parameter is selected, the following optimizations are performed for Cloud X services:

- **Frame identification**
Downlink Cloud X services, primarily video frames, often come in bursts. Specifically, during the transmission of a single video frame, traffic surges occur intensively; whereas during the intervals between video frames, brief idle periods occur or some non-video packets are transmitted. After this function is enabled, the gNodeB identifies packets that belong to the same video frame for Cloud X bearers based on information such as the downlink packet size and packet interval. See [Figure 4-2](#).

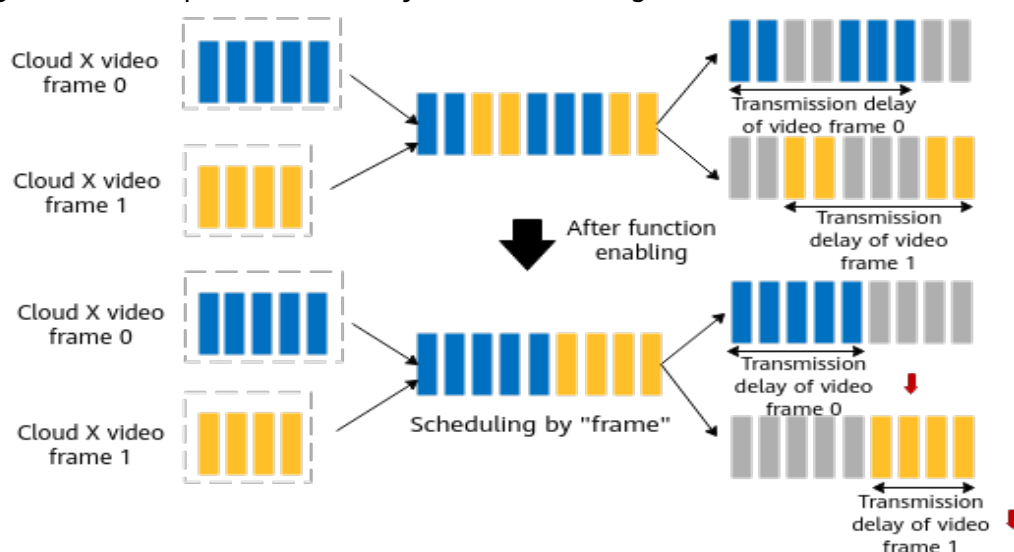
Figure 4-2 Frame identification principle



This function involves the following parameter settings:

- **Frame Large Packet Length Thld:** specified by the **gNBQciBearer.FrameLargePacketLengthThld** parameter. If the length of a Cloud X service packet is greater than this threshold, the packet is considered as a large packet. If the length of a Cloud X service packet is less than or equal to this threshold, the packet is considered as a small packet.
- **Inter-Frame Packet Interval Thld:** specified by the **gNBQciBearer.InterFramePktIntervalThld** parameter. A Cloud X service packet is considered to be the start of a new video frame only when the packet is a large packet and the interval between this packet and the previous large packet is greater than this threshold.
- **Intra-Frame Packet Interval Thld:** specified by the **gNBQciBearer.IntraFramePktIntervalThld** parameter. If the interval between a Cloud X service packet and the previous large packet is less than or equal to this threshold, the two packets are considered to belong to the same video frame.
- **Frame delay-based scheduling**
 In frame delay-based scheduling, packets are scheduled based on frame identification and by frame. See [Figure 4-3](#). This significantly reduces the transmission delay of Cloud X services. Specifically, for the same frame, the gNodeB calculates the frame waiting delay, the existing transmission delay, and predicts the remaining allowed transmission duration within the maximum allowed frame transmission duration threshold. A shorter remaining allowed transmission duration results in a higher packet scheduling priority. The maximum allowed frame transmission duration threshold is specified by the **NRDUCellQciBearer.MaxCloudServSchTransTime** parameter.

Figure 4-3 Principle of frame delay-based scheduling



The **NRDUCellDisch.CloudXResCtrlDlPrbThld** parameter specifies the downlink PRB usage threshold for Cloud X resource control. This parameter is used to prevent Cloud X UEs from occupying excessive PRB resources and causing significant impact on common eMBB UEs.

- When the downlink PRB usage of Cloud X UEs is less than this threshold, frame delay-based scheduling takes effect for all Cloud X UEs.
- When the downlink PRB usage of Cloud X UEs is greater than or equal to this threshold, frame delay-based scheduling no longer takes effect for any Cloud X UE.

In NSA networking, if the Cloud X bearer is configured as an SCG split bearer, the uplink and downlink data of the bearer is transmitted only in the SCG to guarantee the transmission delay of Cloud X services.

Collaboration between Cloud X services and other features and functions:

In low-frequency NR TDD scenarios, the Cloud X multi-cell coordination function allows cell coordination features to take effect for Cloud X UEs, improving anti-interference performance and user experience of Cloud X UEs in overlapping areas. It is recommended that the **INTRA_GNB_SU_COMM_ENH_SW** option of the **NRDUCellCollabServ.MultiCellMimoSwitch** parameter be selected to increase the proportion of cell coordination that takes effect for Cloud X UEs. For details, see *Co-MM (Low-Frequency TDD)*.

4.1.2.2 Cloud X High-Priority UE Guarantee

Cloud X high-priority UE guarantee is introduced to increase the number of Cloud X UEs that meet the service delay requirements in a cell. After this function is enabled, the gNodeB identifies all Cloud X UEs and preferentially guarantees Cloud X UEs identified as high priority.

This function is controlled by the **CLOUD_X_HIGH_PRI_UE_GTEE_SW** option of the **NRDUCellAlgoSwitch.CloudVrSchSwitch** parameter and the following parameters are also involved in this function:

- **NRDUCellDLSch.CloudXResCtrlDlPrbThld**: specifies the downlink PRB usage threshold for Cloud X resource control.
- **gNBQciBearer.CloudXFrmLossDelayThld**: specifies the frame loss delay threshold for Cloud X services. If the transmission delay of a frame of a Cloud X UE is greater than this threshold, the frame is considered as a lost frame. The gNodeB determines the frame loss rate of UEs based on the parameter value and the actual frame transmission delay.

This function is implemented as follows:

1. The gNodeB determines whether a UE is a high-priority Cloud X UE or a common Cloud X UE based on periodic measurements of the following indicators: the total downlink PRB usage of high-priority Cloud X UEs in a cell and the frame loss rate of the Cloud X UE. [Table 4-1](#) describes the conditions for determining the Cloud X UE type.

Table 4-1 Conditions for determining the Cloud X UE type and results

Whether the Total Downlink PRB Usage of High-Priority Cloud X UEs Is Less than the Downlink PRB Usage Threshold for Cloud X Resource Control	Frame Loss Rate of a Cloud X UE	Result
Yes	Low	The UE is identified as a high-priority Cloud X UE.
Yes	High	The UE is identified as a common Cloud X UE.
No	-	Some high-priority Cloud X UEs in the cell are degraded as common Cloud X UEs, ensuring that the total downlink PRB usage of all high-priority Cloud X UEs is less than the downlink PRB usage threshold for Cloud X resource control.

2. Frame delay-based scheduling takes effect only for high-priority Cloud X UEs in the cell. For details about frame delay-based scheduling, see [frame delay-based scheduling](#).

4.1.2.3 Predicted Uplink Scheduling for Cloud X Services

Cloud X services have high requirements on delay. The following optimization measures are taken for Cloud X UEs in the uplink:

- Uplink scheduling:
The periodic buffer status report (BSR) timer is always set to a shorter value, that is, SF1, for Cloud X UEs. This allows the gNodeB to obtain the amount of

data to be scheduled for UEs more promptly and quickly allocate uplink resources, thereby reducing the transmission delay of Cloud X services.

- Smart preallocation is enabled in the uplink:
 - If the **CLOUD_VR_UL_SCH_ENH_SWITCH** option of the **NRDUCELLALGOSWITCH.CloudVrSchSwitch** parameter is deselected, the preallocation algorithm with fixed configuration is used for uplink bearers of identified Cloud X services.
 - If the **CLOUD_VR_UL_SCH_ENH_SWITCH** option of the **NRDUCELLALGOSWITCH.CloudVrSchSwitch** parameter is selected, the gNodeB uses the preallocation algorithm that is based on uplink traffic prediction for bearers carrying Cloud X services, to reduce the uplink padding rate.

4.1.2.4 QCI-specific Scheduling Based on the Downlink Target IBLER

Cloud X services support QCI-specific scheduling based on the downlink target IBLER. This function is controlled by the **DL_IBLER_QCI_SW** option of the **gNBDUSchParamGroup.ScheduleOptSwitch** parameter. After this function is enabled, the **gNBDUSchParamGroup.DITargetIbler** parameter is configured so that services with QCI-specific bearers converge on the target IBLER through outer loop link adaptation (OLLA). Smaller target IBLERs can be configured to reduce the BLER, number of retransmissions, and delay over the air interface. For details about this function, see *Service-differentiated Experience Guarantee*.

NOTE

When multiple bearers are set up for a UE and different QCI-specific bearers have different priorities, the configuration of the switch of QCI-specific scheduling based on the downlink target IBLER (the **DL_IBLER_QCI_SW** option of the **gNBDUSchParamGroup.ScheduleOptSwitch** parameter) corresponding to the QCI with the highest priority will take effect for the UE. To ensure that this function can take effect, the priorities of Cloud X bearers (specified by the **gNBQciBearer.PriorityLevel** parameter) must be higher than those of other bearers.

4.1.2.5 Downlink MCS Adjustment Optimization

The MCS adjustment in downlink adaptation is based on the HARQ feedback result. However, the proportion of ACKs in the HARQ feedback result of small packet services increases after MCS index reduction. For this reason, the MCS adjusted based on the HARQ feedback result cannot match the current channel quality. Downlink MCS adjustment optimization is introduced in Cloud X scenarios. This function is controlled by the **DL_MCS_ADJ_OPT_SW** option of the **NRDUCellDIAMc.DIAMcAlgoSw** parameter. After this option is selected, MCS index reduction is triggered for small-packet services and the step is adaptively adjusted based on IBLER convergence to obtain precise MCS indexes for UEs.

In NR TDD scenarios, this function is supported only in cells with 32T32R or above.

4.1.2.6 Explicit Scheduling for Downlink Second Retransmission

Currently, there are two methods of instructing the transport block size (TBS) used for downlink HARQ retransmissions: implicit scheduling and explicit scheduling.

- Implicit scheduling for HARQ retransmissions: The DCI used for retransmissions does not indicate a specific MCS index. This method is

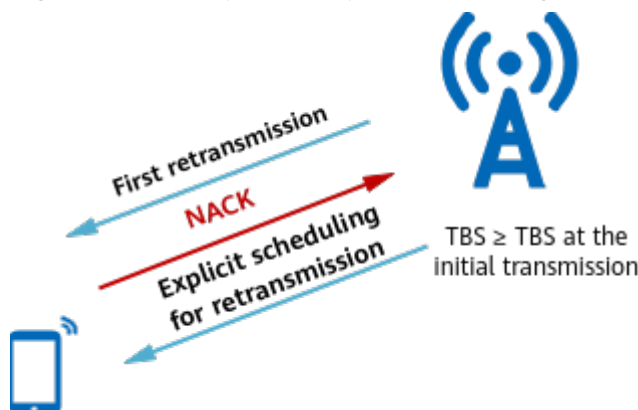
commonly used in a scenario where a UE feeds back a NACK to trigger scheduling for retransmission. In this scenario, the UE has received downlink data, that is, the UE knows the TBS of the data to be transmitted, and the TBS for retransmission is required to be the same as that for initial transmission. Therefore, the UE can calculate the MCS index based on the TBS and the number of RBs.

- Explicit scheduling for HARQ retransmissions: The DCI used for retransmissions indicates a specific MCS index. This method is commonly used in a scenario where a UE feeds back a DTX to trigger scheduling for retransmission. In this scenario, the UE does not receive downlink data and does not know the TBS. As a result, the MCS index cannot be calculated. Therefore, the DCI used for retransmission needs to indicate a specific MCS to the UE.

If the UE does not receive the DCI result for PDSCH scheduling, it sends DTX information to the gNodeB. If the gNodeB mistakenly considers a DTX as a NACK, implicit scheduling for downlink HARQ retransmissions is used. In this case, consecutive retransmission failures may occur, causing packet losses. To avoid this problem and retain the combining gains of the first retransmission, explicit scheduling for downlink second retransmission is introduced.

When the base station detects that the first retransmission result reported by the UE is NACK, this function optimizes the second retransmission mechanism. After this function is enabled, if the TBS that can be transmitted at the second retransmission is greater than or equal to the TBS that can be transmitted at the initial transmission, the base station uses explicit scheduling for retransmission. See [Figure 4-4](#). That is, the base station sends the DCI used for retransmission to the UE to indicate a specific MCS index. This increases the demodulation success rate of the UE and therefore improves the downlink transmission reliability. This function is controlled by the **DL_SECOND_RETRANS_EXPLICIT_SW** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter.

Figure 4-4 Principles of explicit scheduling for downlink second retransmission



NOTE

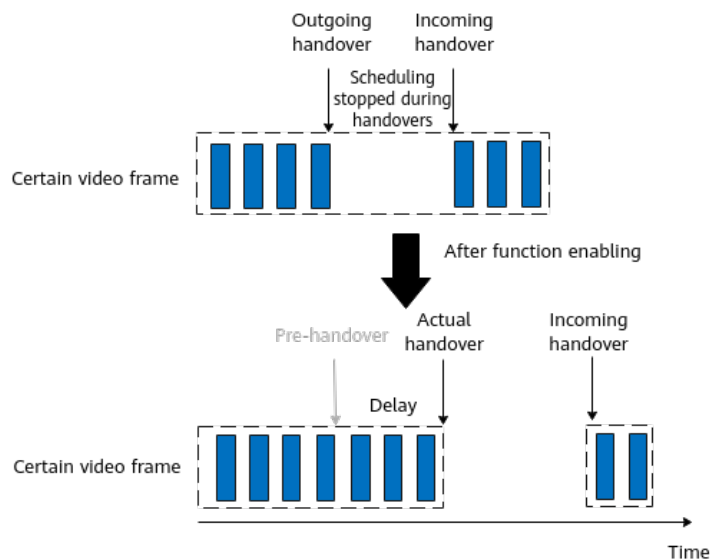
When multiple bearers are set up for a UE and bearers with different QCIs correspond to different priorities, the setting of the **DL_SECOND_RETRANS_EXPLICIT_SW** option for the QCI with the highest priority takes effect. Therefore, to make this function take effect, the priority of the Cloud X bearer (specified by the **gNBQciBearer.PriorityLevel** parameter) must be higher than that of other bearers.

4.1.3 Mobility Management

4.1.3.1 Integrity of Cloud X Frame Transmission During Handovers

The function of "integrity of Cloud X frame transmission during handovers" is introduced for a Cloud X UE to prevent interruptions during data transmission of a frame due to handovers, thereby improving user experience. This function is controlled by the **CLOUD_X_FRAME_INTEGRITY_HO_SW** option of the **NRCellAlgoSwitch.CloudVrSwitch** parameter. After this function is enabled, when a Cloud X UE initiates a handover during the transmission of a frame, the gNodeB postpones the handover until the scheduling of the frame is complete. See [Figure 4-5](#). This prevents data transmission interruptions caused by a handover and increases the probability of Cloud X frame transmission integrity.

Figure 4-5 Principle of integrity of Cloud X frame transmission during handovers



4.1.4 Carrier Management

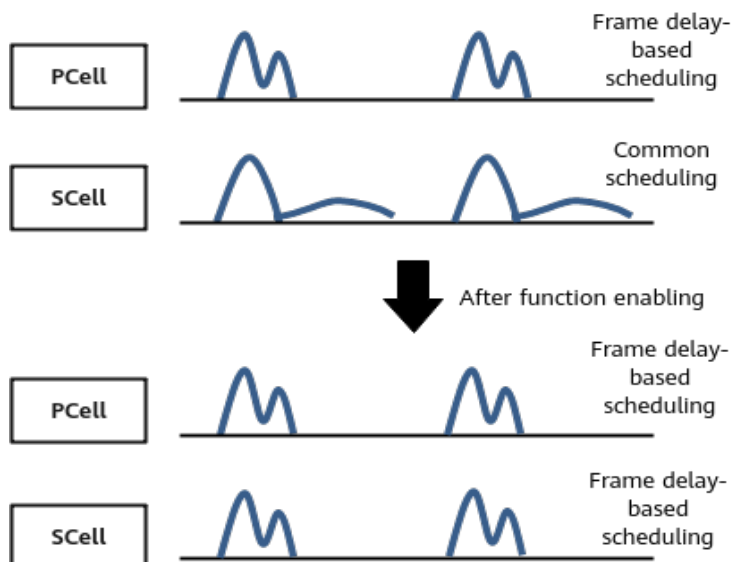
4.1.4.1 Intra-Base-Station CA Downlink Data Split for Cloud X Services

Cloud X services have stringent requirements on delay. However, in the existing CA data split architecture, data cannot be scheduled in a timely manner to meet Cloud X user experience requirements. Therefore, intra-base-station CA downlink data split for Cloud X services is introduced. This function is controlled by the **CLOUD_X_INTRA_GNB_CA_ENH_SW** option of the **NRDUCellAlgoSwitch.CloudVrSchSwitch** parameter.

When the CA activation threshold is reached, CA data split is started. (For details about the threshold, see sections about the SCell configuration procedure in *Carrier Aggregation*.) This function enables SCells to use the same frame delay-based scheduling as the PCell. In this way, distributed data can be scheduled in SCells in a timely manner, realizing complete frame transmission. See [Figure 4-6](#).

(For details about frame delay-based scheduling, see [frame delay-based scheduling](#).)

Figure 4-6 Principle of intra-base-station CA downlink data split for Cloud X services



This function takes effect only when all of the following conditions are met:

- The **CLOUD_X_INTRA_GNB_CA_ENH_SW** option of the **NRDUCellAlgoSwitch.CloudVrSchSwitch** parameter is selected for both the PCell and SCells.
- The PRB usage of Cloud X UEs in an SCell is less than the PRB usage threshold for data split of Cloud X services which is specified by the **NRDUCellIDSch.CloudServSplitPrbThld** parameter.
- The frame scheduling duration of Cloud X UEs is less than the product of the data split duration proportion of Cloud X services (specified by the **gNBQciBearer.CloudServSplitDurProp** parameter) and the maximum transmission duration of a Cloud X frame (specified by the **NRDUCellQciBearer.MaxCloudServSchTransTime** parameter).
- The amount of data to be transmitted for a Cloud X UE is greater than or equal to the product of the Cloud X service data split coefficient (specified by the **gNBQciBearer.CloudServSplitCoef** parameter) and the amount of data that can be scheduled per slot.

4.1.4.2 Latency-based Optimal Carrier Selection

The latency-based optimal carrier selection function is introduced to reduce the downlink air interface delay of Cloud X services. With this function, carriers with the optimal air interface delay are selected for Cloud X UEs based on factors such as coverage, air interface delay, bandwidth, and load. For details about this function, see descriptions about the latency-based optimal carrier selection in *Multi-Frequency Smart Selection*.

4.1.5 Other Functions

4.1.5.1 Uplink ROHC-TCP Compression

Cloud X supports uplink robust header compression (ROHC). After this function is enabled, the header load of uplink data packets is decreased, thereby reducing the amount of uplink data transmitted over the air interface and the bit error rate and delay of radio links. This improves the downlink throughput and service coverage of Cloud X UEs.

This function is controlled by the **ROHC_SW** option of the **NRCCellAlgoSwitch.RohcSwitch** parameter. When the **ROHC_SW** option of the **NRCCellAlgoSwitch.RohcSwitch** parameter is selected and the **UL_PROFILE0X0006** option of the **NRCCellAlgoSwitch.RohcProfiles** parameter is selected, uplink ROHC-TCP compression takes effect.

This function applies only to services whose RLC mode is AM in SA networking. It does not apply to QCI 1 bearers for voice services, QCI 65 and QCI 66 bearers for push to talk (PTT) services, or other extended QCI bearers for PTT services.

After this function is enabled, the gNodeB starts the ROHC procedure when performing Cloud X services.

1. The gNodeB checks whether the UE supports uplink ROHC-TCP compression.
2. If it is supported by the UE, the gNodeB checks whether the UE has entered the ROHC-TCP state. If the UE meets related conditions, the gNodeB instructs the UE to enable this function.
3. After the function takes effect, the UE performs ROHC on headers of uplink TCP/IP service data packets and the gNodeB performs decompression.

4.2 Network Analysis

4.2.1 Benefits

After Cloud X service experience improvement is enabled, the delay perceived by Cloud X UEs is shortened and user experience of such UEs is improved.

After intra-base-station CA downlink data split for Cloud X services is enabled, the user-perceived delay decreases and user experience improves for Cloud X UEs. The gains depend on the following factors:

- Number of carriers: A larger number of carriers on the network and a larger bandwidth of each carrier result in greater benefits.
- Network load: When the network is neither lightly-loaded nor heavily-loaded and the load of SCells remains unchanged, a heavier load of the PCell leads to greater benefits. When the load of the PCell remains unchanged, a heavier load of SCells leads to smaller benefits.
- Coverage: A better coverage of SCells than that of the PCell brings greater benefits.

Explicit scheduling for downlink second retransmission is recommended when the SSB RSRP is less than -100 dBm and the downlink PRB usage of neighboring cells

(indicated by **Downlink Resource Block Utilizing Rate (DU)**) is greater than 50%. After this function is enabled, the perceived delay is shortened, improving user experience of Cloud X UEs. The benefits provided by this function are related to the downlink SSB RSRP and the downlink PRB usage of neighboring cells. A smaller downlink SSB RSRP and a larger downlink PRB usage of neighboring cells indicate higher benefits.

For details about the benefits provided by the latency-based optimal carrier selection function, see descriptions about this function in *Multi-Frequency Smart Selection*.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

- Intra-base-station CA downlink data split for Cloud X services
 - After this function is enabled, the downlink PRB usage (indicated by **Downlink Resource Block Utilizing Rate (DU)**) of the PCell slightly decreases and the downlink PRB usage (indicated by **Downlink Resource Block Utilizing Rate (DU)**) of the SCells slightly increases following data split.
 - After this function is enabled, when the number of Cloud X UEs increases, more downlink resources will be occupied, and the available resources for other UEs will decrease. As a result, the throughput of other eMBB services decreases.
- Explicit scheduling for downlink second retransmission
None
- Latency-based optimal carrier selection
For details, see descriptions about the latency-based optimal carrier selection in *Multi-Frequency Smart Selection*.
- Other functions
After this function is enabled, when the number of Cloud X UEs increases, more downlink resources will be occupied, and the available resources for other UEs will decrease. As a result, the throughput of other eMBB services decreases.

4.3 Requirements

4.3.1 Licenses

- For functions other than QCI-specific scheduling based on the downlink target IBLER, downlink MCS adjustment optimization, and latency-based optimal carrier selection:

Feature ID	Feature Name	Model	Sales Unit
FOFD-081207	Deterministic Latency Guarantee for CloudX	NR0S00UCVE00	per Cell

- QCI-specific scheduling based on the downlink target IBLER
To enable QCI-specific scheduling based on the downlink target IBLER, you need to purchase the license for the FOFD-050208 Service-Differentiated Low Latency feature. The model is NR0S00PSCH00. For details, see *Service-differentiated Experience Guarantee*.
- Downlink MCS adjustment optimization
For low-frequency TDD massive MIMO cells:

Feature ID	Feature Name	Model	Sales Unit
FOFD-061201	AHR Experience Turbo 2.0	NR0S00AET200	per Cell

- Latency-based optimal carrier selection
For details, see descriptions about the latency-based optimal carrier selection in *Multi-Frequency Smart Selection*.

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

- Cloud X high-priority UE guarantee

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cloud X services	CLOUD_VR_SERVICE_SWITCH option of the NRCellAlgoSwitch.CloudVrSwitch parameter	<i>Ultimate Cloud X Experience</i>	Cloud X high-priority UE guarantee depends on Cloud X services or Layered QoS ^a .
a: For details about Layered QoS, see <i>Layered QoS</i> .				

- Predicted uplink scheduling for Cloud X services

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cloud X services	CLOUD_VR_SERVICE_SWITCH option of the NRCellAlgoSwitch.CloudVrSwitch parameter	<i>Ultimate Cloud X Experience</i>	Predicted uplink scheduling for Cloud X services depends on Cloud X services.

- QCI-specific scheduling based on the downlink target IBLER
For details, see descriptions about the QCI-based scheduling based on the target IBLER in *Service-differentiated Experience Guarantee*.
- Downlink MCS adjustment optimization

RAT	Function Name	Function Switch	Reference	Description
TDD	AHR Experience Turbo 2.0	AHR_EXP_TURBO_PHASE2_SW option of the NRDUCellFeatureSw.AhrSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	For low-frequency TDD massive MIMO cells, downlink MCS adjustment optimization takes effect only when AHR Experience Turbo 2.0 is enabled.

- Explicit scheduling for downlink second retransmission

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cloud X services	CLOUD_VR_SERVICE_SWITCH option of the NRCellAlgoSwitch.CloudVrSwitch parameter	<i>Ultimate Cloud X Experience</i>	The explicit scheduling for downlink second retransmission function depends on the Cloud X services or service-differentiated link reliability ^a .
a: For Cloud X services, explicit scheduling for downlink second retransmission depends on Cloud X services. For non-Cloud X services, explicit scheduling for downlink second retransmission takes effect only when service-differentiated link reliability is enabled (controlled by the LINK_RELIABILITY_DIFF_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter). For details, see the section "Explicit Scheduling for Downlink Second Retransmission" in <i>Service-differentiated Experience Guarantee</i>.				

- Integrity of Cloud X frame transmission during handovers

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cloud X UE identification	gNBQciBearer.ServiceType set to EMBB_CLOUD_VR or EMBB_CLOUD_GAMING	<i>Ultimate Cloud X Experience</i>	The integrity of Cloud X frame transmission during handovers function takes effect only when Cloud X UE identification is enabled.

- Intra-base-station CA downlink data split for Cloud X services

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-gNodeB intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Intra-base-station CA downlink data split for Cloud X services depends on downlink intra-gNodeB intra-band CA.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-gNodeB intra-FR inter-band CA	INTRA_FR_IN TER_BAND_C A_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Intra-base-station CA downlink data split for Cloud X services depends on downlink intra-gNodeB intra-FR inter-band CA.
Low-frequency TDD	Cloud X services	CLOUD_VR_SERVICE_SWITCH option of the NRCCellAlgoSwitch.CloudVrSwitch parameter	<i>Ultimate Cloud X Experience</i>	Intra-base-station CA downlink data split for Cloud X services requires that Cloud X services be enabled.

- Latency-based optimal carrier selection
For details, see descriptions about the latency-based optimal carrier selection in *Multi-Frequency Smart Selection*.
- Other functions
None

4.3.2.2 Mutually Exclusive Functions

- Cloud X UE identification

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink PDU Set integrity protection	INTEGRITY_GUARANTEE_ALGO_SW option of the gNBQciBearer.DLLayeredQosSwitch parameter and CLOUD_X_HIGH_PRI_UE_GUARANTEE_SW option of the NRDUCellAlgoSwitch.CloudVrSchSwitch parameter	<i>Layered QoS</i>	When Cloud X UE identification is enabled, the INTEGRITY_GUARANTEE_ALGO_SW option of the gNBQciBearer.DLLayeredQosSwitch parameter cannot be selected.

- QCI-specific scheduling based on the downlink target IBLER
For details, see descriptions about the QCI-based scheduling based on the target IBLER in *Service-differentiated Experience Guarantee*.
- Latency-based optimal carrier selection
For details, see descriptions about the latency-based optimal carrier selection in *Multi-Frequency Smart Selection*.
- Other functions
None

4.3.2.3 Function Impacts

- Cloud X UE identification

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>UE Power Saving</i>	After Cloud X UE identification and SSSG switching are enabled, UEs performing Cloud X services will not be switched to an SSSG.
Low-frequency TDD	RedCap SSSG switching	REDCAP_SSSG_SWITCHING_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter or REDCAP_SSSG_SWITCH_EXTENSION_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>RedCap</i>	After Cloud X UE identification and RedCap SSSG switching are enabled, UEs performing Cloud X services will not be switched to an SSSG.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Flexible spectrum access	FLEX_SPCT_ACCESS_SWITCH option of the NRDUCellCarrierMgmt.MultiBandFlexSwitch parameter	<i>Uplink Flexible Spectrum Scheduling</i>	Flexible spectrum access and Cloud X UE identification cannot take effect at the same time.
Low-frequency TDD	Time-domain power saving BWP	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	UEs with Cloud X services in effect do not switch to BWP2.

- Cloud X services

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink data compression	• UDC_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter • UPLINK_SIG_COMPRESS_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter • QCI_UDC_SW option of the gNBQciBearer.PdcpAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Cloud X services and uplink data compression cannot take effect simultaneously for the same bearer of a UE. Cloud X services preferentially take effect for the bearer if they are enabled together with uplink data compression.
Low-frequency TDD	Uplink fallback to LTE	UL_FALLBACK_TO_LTE_SWITCH option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>	When uplink fallback to LTE takes effect on the uplink path of a Cloud X bearer, the uplink data transmission changes to the MCG.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink data transmission path selection	• LTE: NSA_DC_UPL_PATH_SELECTION_SW option of the NsaDcMgmtConfig.NsaDcAlgorithmSwitch parameter • NR: NSA_DC_UPL_PATH_SELECTION_SW option of the NRCellNsaDcConfig.NsaDcAlgorithmSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	When uplink data transmission path selection takes effect on the uplink path of a Cloud X bearer, the uplink data transmission changes to the MCG.

- Cloud X high-priority UE guarantee

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink PDU Set integrity protection	INTEGRITY_GUARANTEE_ALGO_SW option of the gNBQciBearer.DLLayeredQoSAlgorithmSwitch parameter and CLOUD_X_HIGH_PRI_UE_GUARANTEE_SW option of the NRDUCellAlgorithmSwitch.CloudVrSchSwitch parameter	<i>Layered QoS</i>	If both Cloud X bearers and non-Cloud X bearers exist in a cell, the downlink PDU Set integrity guarantee function can also take effect for non-Cloud X bearers. This reduces the number of downlink PRBs available for Cloud X bearers, reducing the gains of Cloud X high-priority UE guarantee.

- QCI-specific scheduling based on the downlink target IBLER
For details, see descriptions about the QCI-based scheduling based on the target IBLER in *Service-differentiated Experience Guarantee*.
- Intra-base-station CA downlink data split for Cloud X services

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrierMgmt.CAdvancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	CA Performance Boost Based on Multi-dimensional Optimization aims to improve the UE scheduling efficiency and therefore increase the average downlink throughput of CA UEs. Intra-base-station CA downlink data split for Cloud X services aims to ensure the delay and user experience of Cloud X UEs after data split. Therefore, when CA Performance Boost Based on Multi-dimensional Optimization takes effect, the benefits provided by intra-base-station CA downlink data split for Cloud X services decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed data volume-triggered SCell deactivation	SPLIT_VOL_BASED_SCELL_DEACT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Carrier Aggregation</i>	When both intra-base-station CA downlink data split for Cloud X services and distributed data volume-triggered SCell deactivation are enabled: • A larger value of the gNBQciBearer.CloudServSplitCoef parameter results in a higher CA downlink data split threshold for Cloud X UEs. If the buffered data volume is lower than the data split threshold for a long time, SCells will be deactivated and the values of N.CA.SCell.Deact.Att and N.CA.SCell.Deact.Succ counters will increase. • A smaller value of the gNBQciBearer.CloudServSplitCoef parameter results in a lower CA downlink data split threshold, a longer SCell-activated duration, and higher UE power consumption for Cloud X UEs.

- Latency-based optimal carrier selection
For details, see descriptions about the latency-based optimal carrier selection in *Multi-Frequency Smart Selection*.
- Other functions
None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

- Downlink MCS adjustment optimization
For TDD, all NR-capable RF modules with at least 32T32R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- Latency-based optimal carrier selection
For details, see descriptions about the latency-based optimal carrier selection in *Multi-Frequency Smart Selection*.
- Other functions
All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

- Downlink MCS adjustment optimization
In NR TDD scenarios, this function is supported only in cells with 32T32R or above.
- Latency-based optimal carrier selection
For details, see descriptions about the latency-based optimal carrier selection in *Multi-Frequency Smart Selection*.
- Other functions
None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment. For details about the data

preparation for the latency-based optimal carrier selection function, see descriptions about this function in *Multi-Frequency Smart Selection*.

Table 4-2 Parameters used for activation

RAT	Parameter Name	Parameter ID	Setting Notes
TDD	Service Type	gNBQciBearer.ServiceType	This parameter specifies the QCI type of Cloud X UEs. You are advised to set this parameter to EMBB_CLOUD_VR or EMBB_CLOUD_GAMING based on the service type.
TDD	CloudX Switch	NRCCellAlgoSwitch.CloudVrSwitch	It is recommended that the CLOUD_VR_SERVICE_SWITCH option be selected to enable Cloud X services.
TDD	Frame Large Packet Length Thld	gNBQciBearer.FrameLargePacketLengthThld	It is recommended that this parameter be set to 1300 when the CLOUD_VR_SERVICE_SWITCH option of the NRCCellAlgoSwitch.CloudVrSwitch parameter is selected.
TDD	Inter-Frame Packet Interval Thld	gNBQciBearer.InterFramePktIntervalThld	It is recommended that this parameter be set to 5 when the CLOUD_VR_SERVICE_SWITCH option of the NRCCellAlgoSwitch.CloudVrSwitch parameter is selected.
TDD	Intra-Frame Packet Interval Thld	gNBQciBearer.IntraFramePktIntervalThld	It is recommended that this parameter be set to 2 when the CLOUD_VR_SERVICE_SWITCH option of the NRCCellAlgoSwitch.CloudVrSwitch parameter is selected.

RAT	Parameter Name	Parameter ID	Setting Notes
TDD	CloudX Sch Switch	NRDUCellAlgoSwitch.<i>CloudVrSchSwitch</i>	It is recommended that the CLOUD_VR_UL_SCH_ENH_SWITCH option be selected to enable predicted uplink scheduling for Cloud X services. It is recommended that the CLOUD_X_HIGH_PRI_UE_GTEE_SW option be selected when there are a large number of Cloud X UEs to implement Cloud X high-priority UE guarantee.
TDD	ROHC Switch	NRCCellAlgoSwitch.<i>RoHcSwitch</i>	To enable uplink ROHC-TCP compression, select the ROHC_SW option.
TDD	ROHC Profiles	NRCCellAlgoSwitch.<i>RoHcProfiles</i>	To enable uplink ROHC-TCP compression, select the UL_PROFILE0X0006 option.
TDD	CloudX Resource Control DL PRB Usage Threshold	NRDUCellDISch.<i>CloudXResCtrlDLPrbThld</i>	It is recommended that this parameter be set to 80 when Cloud X services are enabled. When Cloud X high-priority UE guarantee is enabled and the corresponding cell is configured with a dedicated Cloud X network slice, it is recommended that this parameter be set to the value of the NRDUCellRes.<i>DLRbAverageRatio</i> or NRDUCellRes.<i>DLRbMinRatio</i> parameter of the network slice. For details about the parameter values, see <i>Network Slicing</i>.

RAT	Parameter Name	Parameter ID	Setting Notes
TDD	CloudX Frame Loss Delay Thld	gNBQciBearer.CloudXFrmLossDelayThld	When Cloud X high-priority UE guarantee is enabled: • 10 for Cloud VR services • 15 for Cloud Gaming services
TDD	Max Cloud Service Scheduling Transfer Time	NRDUCellQciBearer.MaxCloudServSchTransTime	It is recommended that this parameter be set to 8 if the CLOUD_X_HIGH_PRI_UE_GTEE_SW option of the NRDUCellAlgoSwitch.CloudVrSchSwitch parameter is deselected. It is recommended that this parameter be set to 10 if the CLOUD_X_HIGH_PRI_UE_GTEE_SW option of the NRDUCellAlgoSwitch.CloudVrSchSwitch parameter is selected.
Low-frequency TDD	Schedule Option Switch	gNBDUSchParamGroup.ScheduleOptSwitch	To enable QCI-specific scheduling based on the downlink target IBLER, select the DL_IBLER_QCI_SW option.
Low-frequency TDD	Low Latency Switch	NRDUCellAlgoSwitch.LowLatencySwitch	When enabling QCI-specific scheduling based on the downlink target IBLER, select the SERV_DIFF_LOW_LATENCY_SW option.
Low-frequency TDD	Downlink Target IBLER	gNBDUSchParamGroup.DlTargetIbler	The value 5000 is recommended when QCI-specific scheduling based on the downlink target IBLER is enabled. A larger value of this parameter results in a larger IBLER of Cloud X UEs in the cell and a longer user-perceived delay.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Priority Level	gNBQciBearer.Priorit yLevel	The value 7 is recommended when QCI-specific scheduling based on the downlink target IBLER is enabled.
Low-frequency TDD	CloudX Switch	NRCellAlgoSwitch.Cl oudVrSwitch	To enable integrity of Cloud X frame transmission during handovers, select the CLOUD_X_FRAME_INTEG RITY_HO_SW option.
Low-frequency TDD	CloudX Sch Switch	NRDUCellAlgoSwitch .CloudVrSchSwitch	To enable intra-base-station CA downlink data split for Cloud X services, select the CLOUD_X_INTRA_GNB_C A_ENH_SW option.
Low-frequency TDD	Cloud Service Split PRB Usage Threshold	NRDUCellDLSch.Clou dServSplitPrbThld	This parameter must be set when intra-base-station CA downlink data split for Cloud X services is required. Use the recommended value.
Low-frequency TDD	Cloud Service Split Duration Proportion	gNBQciBearer.Cloud ServSplitDurProp	This parameter must be set when intra-base-station CA downlink data split for Cloud X services is required. Use the recommended value.
Low-frequency TDD	Cloud Service Split Coefficient	gNBQciBearer.Cloud ServSplitCoef	This parameter must be set when intra-base-station CA downlink data split for Cloud X services is required. Use the recommended value.
Low-frequency TDD	Downlink AMC Algorithm Switch	NRDUCellDLAmc.DLA mcAlgoSw	To enable downlink MCS adjustment optimization, select the DL_MCS_ADJ_OPT_SW option.
Low-frequency TDD	Service Differentiation Algorithm Switch	NRDUCellQciBear- er.ServiceDiffAlgoS- witch	To enable explicit scheduling for downlink second retransmission, select the DL_SECOND_RETRANS_E XPPLICIT_SW option.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Priority Level	gNBQciBearer.Priorit yLevel	To enable explicit scheduling for downlink second retransmission, it is recommended that this parameter be set to 7 .

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Cloud X UE identification (The following is an example of Cloud X bearer services, which need to be configured based on actual QCI.)
//Configuring bearers for Cloud VR services
MOD GNBQCIBEARER: Qci=6, ServiceType=EMBB_CLOUD_VR;
//Configuring bearers for Cloud Gaming services
MOD GNBQCIBEARER: Qci=10, ServiceType=EMBB_CLOUD_GAMING;
- Cloud X services
//Turning on the Cloud X service switch
MOD NRCELLALGOSWITCH: NrCellId=1, CloudVrSwitch=CLOUD_VR_SERVICE_SWITCH-1;
//Configuring threshold parameters related to frame identification (using Cloud VR services as an example)
MOD GNBQCIBEARER: Qci=6, FrameLargePacketLengthThld=1300, InterFramePktIntervalThld=5, IntraFramePktIntervalThld=2;
//Configuring the downlink PRB usage threshold for Cloud X resource control
MOD NRDUCELLDLSCH: NrDuCellId=1, CloudXResCtrlDlPrbThld=80;
//Configuring Max CloudX Scheduling Transfer Time (using Cloud VR services as an example)
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=6, MaxCloudServSchTransTime=10;
- Cloud X high-priority UE guarantee
//Turning on the switch for Cloud X high-priority UE guarantee
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CloudVrSchSwitch=CLOUD_X_HIGH_PRI_UE_GTEE_SW-1;
//Configuring the frame loss delay threshold for Cloud VR services
MOD GNBQCIBEARER: Qci=6, CloudXFrmlLossDelayThld=10;
//Configuring the frame loss delay threshold for Cloud Gaming services
MOD GNBQCIBEARER: Qci=10, CloudXFrmlLossDelayThld=15;
- Predicted uplink scheduling for Cloud X services
//Turning on the switch for predicted uplink scheduling for Cloud X services
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CloudVrSchSwitch=CLOUD_VR_UL_SCH_ENH_SWITCH-1;
- QCI-specific scheduling based on the downlink target IBLER
//Turning on the switch for service-differentiated low latency
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, LowLatencySwitch=SERV_DIFF_LOW_LATENCY_SW-1;
//Configuring a scheduling parameter group, turning on the switch for QCI-specific scheduling based on the downlink target IBLER, and setting IblerTarget to 5000

```
ADD GNBUSCHPARAMGROUP: ScheduleParamGroupId=9, ScheduleOptSwitch=DL_IBLER_QCI_SW-1,
DLTargetIbler=5000;
//Binding the QCI to the corresponding scheduling parameter group
MOD NRDUCELLQCI BEARER: NrDuCellId=1, Qci=6, ScheduleParamGroupId=9;
//Configuring the priority of the Cloud X bearer to be higher than that of other bearers
MOD GNBQCI BEARER: Qci=6, PriorityLevel=7;
```

- **Downlink MCS adjustment optimization**
//Turning on the switch for downlink MCS adjustment optimization
MOD NRDUCELLDLAMC: NrDuCellId=1, DLAmcAlgoSw=DL_MCS_ADJ_OPT_SW-1;
- **Explicit scheduling for downlink second retransmission**
//Enabling explicit scheduling for downlink second retransmission for Cloud VR services and setting the priority of Cloud VR services to be higher than that of other bearers
MOD NRDUCELLQCI BEARER: NrDuCellId=1, Qci=6, ServiceDiffAlgoSwitch=DL_SECOND_RETRANS_EXPLICIT_SW-1;
MOD GNBQCI BEARER: Qci=6, PriorityLevel=7;
//Enabling explicit scheduling for downlink second retransmission for Cloud Gaming services and setting the priority of Cloud Gaming services to be higher than that of other bearers
MOD NRDUCELLQCI BEARER: NrDuCellId=1, Qci=10, ServiceDiffAlgoSwitch=DL_SECOND_RETRANS_EXPLICIT_SW-1;
MOD GNBQCI BEARER: Qci=10, PriorityLevel=7;
- **Integrity of Cloud X frame transmission during handovers**
//Turning on the switch for integrity of Cloud X frame transmission during handovers (Before turning on this switch, select the CLOUD_VR_SERVICE_SWITCH option of the NRCellAlgoSwitch.CloudVrSwitch parameter.)
MOD NRCELLALGOSWITCH: NrCellId=1, CloudVrSwitch=CLOUD_X_FRAME_INTEGRITY_HO_SW-1;
- **Intra-base-station CA downlink data split for Cloud X services**
//Turning on the switch for intra-base-station CA downlink data split for Cloud X services (assuming that cell 1 is the PCell and cell 2 is an SCell)
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CloudVrSchSwitch=CLOUD_X_INTRA_GNB_CA_ENH_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=2, CloudVrSchSwitch=CLOUD_X_INTRA_GNB_CA_ENH_SW-1;
//Setting the Cloud Service Split PRB Usage Threshold
MOD NRDUCELLDLSCH: NrDuCellId=2, CloudServSplitPrbThld=50;
//Setting the Cloud Service Split Duration Proportion and Cloud Service Split Coefficient
MOD GNBQCI BEARER: Qci=6, CloudServSplitDurProp=8, CloudServSplitCoef=2;
- **Latency-based optimal carrier selection: For details, see the related descriptions about this function in *Multi-Frequency Smart Selection*.**
- **Uplink ROHC-TCP compression**
//Turning on the ROHC switch
MOD NRCELLALGOSWITCH: NrCellId=1, RohcSwitch=ROHC_SW-1, RohcProfiles=UL_PROFILE0X0006-1;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- **Uplink ROHC-TCP compression**
//Turning off the ROHC switch
MOD NRCELLALGOSWITCH: NrCellId=1, RohcSwitch=ROHC_SW-0, RohcProfiles=UL_PROFILE0X0006-0;
- **Latency-based optimal carrier selection: For details, see the related descriptions about this function in *Multi-Frequency Smart Selection*.**
- **Intra-base-station CA downlink data split for Cloud X services**
//Turning off the switch for intra-base-station CA downlink data split for Cloud X services
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CloudVrSchSwitch=CLOUD_X_INTRA_GNB_CA_ENH_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=2, CloudVrSchSwitch=CLOUD_X_INTRA_GNB_CA_ENH_SW-0;
- **Integrity of Cloud X frame transmission during handovers**

- ```
//Turning off the switch for integrity of Cloud X frame transmission during handovers
MOD NRCELLALGOSWITCH: NrCellId=1, CloudVrSwitch=CLOUD_X_FRAME_INTEGRITY_HO_SW-0;
```
- **Explicit scheduling for downlink second retransmission**  

```
//Disabling explicit scheduling for downlink second retransmission for Cloud VR services
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=6,
ServiceDiffAlgoSwitch=DL_SECOND_RETRANS_EXPLICIT_SW-0;
//Disabling explicit scheduling for downlink second retransmission for Cloud Gaming services
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=10,
ServiceDiffAlgoSwitch=DL_SECOND_RETRANS_EXPLICIT_SW-0;
```
- **Downlink MCS adjustment optimization**  

```
//Turning off the switch for downlink MCS adjustment optimization
MOD NRDUCELLDLAMC: NrDuCellId=1, DLAmcAlgoSw=DL_MCS_ADJ_OPT_SW-0;
```
- **QCI-specific scheduling based on the downlink target IBLER**  

```
//Turning off the switch for QCI-specific scheduling based on the downlink target IBLER
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=9, ScheduleOptSwitch=DL_IBLER_QCI_SW-0;
```
- **Predicted uplink scheduling for Cloud X services**  

```
//Turning off the switch for predicted uplink scheduling for Cloud X services
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
CloudVrSchSwitch=CLOUD_VR_UL_SCH_ENH_SWITCH-0;
```
- **Cloud X high-priority UE guarantee**  

```
//Turning off the switch for Cloud X high-priority UE guarantee
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
CloudVrSchSwitch=CLOUD_X_HIGH_PRI_UE_GTEE_SW-0;
```
- **Cloud X services**  

```
//Turning off the Cloud X service switch (After the Cloud X service switch is turned off, all Cloud X
functions become ineffective. The switch used for deactivation must be the same as that for
activation.)
MOD NRCELLALGOSWITCH: NrCellId=1, CloudVrSwitch=CLOUD_VR_SERVICE_SWITCH-0;
```
- **Cloud X UE identification**  

```
//Changing the service type of Cloud VR bearers to normal service
MOD GNBQCIBEARER: Qci=6, ServiceType=NORMAL;
//Changing the service type of Cloud Gaming bearers to normal service
MOD GNBQCIBEARER: Qci=10, ServiceType=NORMAL;
```

#### 4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

##### NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 4.4.2 Activation Verification

1. Ensure that 5QI (in SA networking) or QCI (in NSA networking) bearers are configured for Cloud X services in the cell. You can run the **LST GNBQCIBEARER** command to check whether the **Service Type** of the 5QI or QCI is **EMBB\_CLOUD\_VR** or **EMBB\_CLOUD\_GAMING**.
2. Run the following MML command to subscribe to the performance counter objects.
  - a. SA scenario (using 5QI 6 as an example)  

```
MOD GNB5QICONFIG: Nr5qi=6, CounterObjSubscriptionInd=YES;
```
  - b. NSA scenario (using QCI 6 as an example)  

```
MOD GNBQCIBEARER: Qci=6, CounterObjSubscriptionInd=YES;
```
3. After the preceding configurations are complete, use the following activation verification methods to check whether the functions have taken effect.



a. SA scenario

- i. Cloud X UE identification: Observe the values of the **N.QosFlow.Max.Cell5QI**, **N.QosFlow.Avg.Cell5QI**, **N.User.RRCCConn.Active.DuCell5QI.Max**, and **N.User.RRCCConn.Active.DuCell5QI.Avg** counters. If the value of any of these counters is not 0, this function has taken effect.
- ii. Uplink ROHC-TCP: Observe the values of the **N.PDCP.UL.ROHC.TotalDecomp.Data**, **N.PDCP.UL.ROHC.Data.AfterDecomp.Bytes**, **N.PDCP.UL.ROHC.Data.BeforeDecomp.Bytes**, and **N.PDCP.UL.ROHC.DecompFail.Data** counters. If the value of any of these counters is not 0, this function has taken effect.
- iii. Predicted uplink scheduling for Cloud X services: Observe the values of the 5QI-specific average uplink delay at the MAC layer and 5QI-specific total uplink delay at the MAC layer. If the value of any of these indicators decreases, this function has taken effect.
- iv. Other functions: Observe the 5QI-specific average downlink delay at the MAC layer and 5QI-specific total downlink delay at the MAC layer. If the value of any of these indicators decreases, the related function has taken effect.

5QI-specific average uplink delay at the MAC layer =  
**N.MAC.UL.Cell5QI.Delay/N.MAC.UL.Packets.Cell5QI**

5QI-specific total uplink delay at the MAC layer:  
**N.MAC.UL.Cell5QI.Delay**

5QI-specific average downlink delay at the MAC layer =  
**N.MAC.DL.Cell5QI.Delay/N.MAC.DL.Packets.Cell5QI**

5QI-specific total downlink delay at the MAC layer:  
**N.MAC.DL.Cell5QI.Delay**

**Table 4-3** Performance counters for activation verification of Cloud X services in SA networking

| Counter ID | Counter Name                                 |
|------------|----------------------------------------------|
| 1911820520 | <b>N.QosFlow.Max.Cell5QI</b>                 |
| 1911820521 | <b>N.QosFlow.Avg.Cell5QI</b>                 |
| 1911820488 | <b>N.User.RRCCConn.Active.DuCell5QI.Max</b>  |
| 1911820489 | <b>N.User.RRCCConn.Active.DuCell5QI.Avg</b>  |
| 1911833754 | <b>N.MAC.DL.Packets.Cell5QI</b>              |
| 1911833757 | <b>N.MAC.DL.Cell5QI.Delay</b>                |
| 1911833764 | <b>N.MAC.UL.Packets.Cell5QI</b>              |
| 1911833761 | <b>N.MAC.UL.Cell5QI.Delay</b>                |
| 1911825080 | <b>N.PDCP.UL.ROHC.TotalDecomp.Data</b>       |
| 1911825081 | <b>N.PDCP.UL.ROHC.Data.AfterDecomp.Bytes</b> |

| Counter ID | Counter Name                                  |
|------------|-----------------------------------------------|
| 1911825082 | <b>N.PDCP.UL.ROHC.Data.BeforeDecomp.Bytes</b> |
| 1911825083 | <b>N.PDCP.UL.ROHC.DecompFail.Data</b>         |

b. NSA scenario

- i. Cloud X UE identification: Observe the values of **N.User.RRCCConn.Active.QCI.Max** and **N.User.RRCCConn.Active.QCI.Avg**. If the value of any of these counters is not 0, this function has taken effect.
- ii. Predicted uplink scheduling for Cloud X services: Observe the values of the QCI-specific average uplink delay at the MAC layer and QCI-specific total uplink delay at the MAC layer. If the value of any of these indicators decreases, this function has taken effect.
- iii. Other NSA-capable functions: Observe the QCI-specific average downlink delay at the MAC layer and QCI-specific total downlink delay at the MAC layer. If the value of any of these indicators decreases, the related function has taken effect.

QCI-specific average uplink delay at the MAC layer =  
**N.MAC.UL.QCI.Delay/N.MAC.UL.Packets.QCI**

QCI-specific total uplink delay at the MAC layer: **N.MAC.UL.QCI.Delay**

QCI-specific average downlink delay at the MAC layer =  
**N.MAC.DL.QCI.Delay/N.MAC.DL.Packets.QCI**

QCI-specific total downlink delay at the MAC layer: **N.MAC.DL.QCI.Delay**

**Table 4-4** Performance counters for activation verification of Cloud X services in NSA networking

| Counter ID | Counter Name                          |
|------------|---------------------------------------|
| 1911817085 | <b>N.User.RRCCConn.Active.QCI.Max</b> |
| 1911817086 | <b>N.User.RRCCConn.Active.QCI.Avg</b> |
| 1911851737 | <b>N.MAC.DL.Packets.QCI</b>           |
| 1911851733 | <b>N.MAC.DL.QCI.Delay</b>             |
| 1911851738 | <b>N.MAC.UL.Packets.QCI</b>           |
| 1911851729 | <b>N.MAC.UL.QCI.Delay</b>             |

## 4.4.3 Network Monitoring

Enabling this function improves video display experience by ensuring video playing rate and reducing the number of video freezes.

- The following methods are recommended for precise performance monitoring:
  - Perform DTs to observe the gains of a single UE: Play and observe the effect of Cloud X services on the UE, record the number of pauses during

the play, and check whether dizziness occurs. Screen recording and other video recording methods can be used to collect and analyze statistics.

- Leverage capabilities of video service KPI evaluation on the core network side or use third-party video service KPI evaluation equipment:  
Obtain video KPIs and MTP delay.
- The following method is recommended for rough performance monitoring:
  - UE throughput: You are advised to monitor experience of a single UE to evaluate the increases in the UE throughput.

# 5 Parameters

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The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

## NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

## **FAQ 1: How do I find the parameters related to a certain feature from parameter reference?**

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

## **FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?**

- Step 1** Open the EXCEL file of the used reserved parameter list.

**Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

# 6 Counters

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The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

## NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

## **FAQ: How do I find the counters related to a certain feature from performance counter reference?**

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

# 7 Glossary

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For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

# 8 Reference Documents

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1. 3GPP TS 38.104: "Base Station (BS) radio transmission and reception (Release 17)"
2. *EPC-based NSA Basics*
3. *EPC-based NSA Performance Enhancement*
4. *Carrier Aggregation*
5. *Network Slicing*
6. *Multi-Frequency Smart Aggregation*
7. *Co-MM (Low-Frequency TDD)*
8. *Massive MIMO AHR (Low-Frequency TDD)*
9. *License Control Item Lists*
10. *Uplink Flexible Spectrum Scheduling*
11. *Multi-Frequency Smart Selection*
12. *Service-differentiated Experience Guarantee*
13. *RedCap*
14. *Layered QoS*
15. *UE Power Saving*
16. *Turbo Uplink (Low-Frequency TDD)*